

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Projector
 Manufacturer: Barco
 Firmware Version: N/A
 Model(s): FS70-4K6

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.3

Version History

Module Version	Date	Notes
1_0_0_0	3/16/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

Focus	'In'	'Out'
['Steps']	'1' – '16'	
Illumination	0 to 100 in steps of 1	
Input	'DVI 1' 'DisplayPort 2' 'SDI'	'DVI 2' 'HDMI' 'DisplayPort 1' 'HDBaseT'
Power	'On'	'Off'
Shutter	'Open'	'Closed'
TestPattern	'On'	'Off'
TestPatternSelect	'Focus Black' 'Checker Board' 'Monoscope'	'Ansi Lumen' 'Focus' 'Outline' 'Scenergix' 'Cross Hatch' 'Color Bars'
Zoom	'In'	'Out'
['Steps']	'1' – '16'	

Status Available:

ConnectionStatus	'Connected'	'Disconnected'
Illumination	0 to 100 in steps of 1	
Input	'DVI 1' 'DisplayPort 2' 'SDI'	'DVI 2' 'HDMI' 'DisplayPort 1' 'HDBaseT'
Power	'On'	'Off'
Shutter	'Open'	'Closed'
TestPattern	'On'	'Off'
TestPatternSelect	'Focus Black' 'Checker Board' 'Monoscope'	'Ansi Lumen' 'Focus' 'Outline' 'Scenergix' 'Cross Hatch' 'Color Bars'

Cable and Adapter Requirements:

Captive Screw to Male DB9 RS-232 Serial Cable

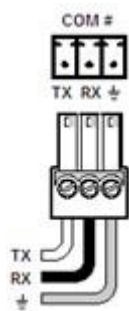
Notes for the Device:

Serial communication:

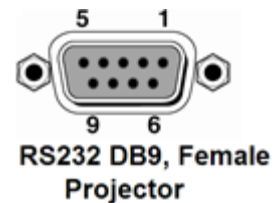
Port Type: RS-232
Baud Rate: 19200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
RxD	←	2	TxD
TxD	→	3	RxD
GND	→	5	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	9090
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Focus In Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepforward", "steps": 1}, "id": 1}\x0D\x0A
Focus In Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepforward", "steps": 16}, "id": 1}\x0D\x0A
Focus Out Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepreverse", "steps": 1}, "id": 1}\x0D\x0A
Focus Out Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.focus.stepreverse", "steps": 16}, "id": 1}\x0D\x0A
Illumination 0	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "illumination.sources.laser.power", "value": 0.0}, "id": 2}\x0D\x0A
Illumination 100	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "illumination.sources.laser.power", "value": 100.0}, "id": 2}\x0D\x0A
Input DisplayPort 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DisplayPort 1"}, "id": 5}\x0D\x0A
Input DisplayPort 2	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DisplayPort 2"}, "id": 5}\x0D\x0A
Input DVI 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DVI 1"}, "id": 5}\x0D\x0A
Input DVI 2	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "DVI 2"}, "id": 5}\x0D\x0A
Input HDBaseT	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "HDBaseT"}, "id": 5}\x0D\x0A
Input HDMI	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "HDMI"}, "id": 5}\x0D\x0A
Input SDI	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.window.main.source", "value": "SDI"}, "id": 5}\x0D\x0A
Power Off	{"jsonrpc": "2.0", "method": "system.poweroff", "params": {}, "id": 3}\x0D\x0A
Power On	{"jsonrpc": "2.0", "method": "system.poweron", "params": {}, "id": 3}\x0D\x0A
Shutter Closed	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.shutter.target", "value": "Closed"}, "id": 4}\x0D\x0A
Shutter Open	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.shutter.target", "value": "Open"}, "id": 4}\x0D\x0A
Test Pattern Off	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.show", "value": false}, "id": 6}\x0D\x0A
Test Pattern On	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.show", "value": true}, "id": 6}\x0D\x0A
Test Pattern Select Ansi Lumen	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Ansi Lumen"}, "id": 7}\x0D\x0A
Test Pattern Select Checker Board	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Checker Board"}, "id": 7}\x0D\x0A
Test Pattern Select Color Bars	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Color Bars"}, "id": 7}\x0D\x0A

Test Pattern Select Cross Hatch	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Cross Hatch"}, "id": 7}\x0D\x0A
Test Pattern Select Focus	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Focus"}, "id": 7}\x0D\x0A
Test Pattern Select Focus Black	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Focus Black"}, "id": 7}\x0D\x0A
Test Pattern Select Monoscope	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Monoscope"}, "id": 7}\x0D\x0A
Test Pattern Select Outline	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Outline"}, "id": 7}\x0D\x0A
Test Pattern Select Scenergix	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "image.testpattern.selected", "value": "internal:Scenergix"}, "id": 7}\x0D\x0A
Zoom In Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepforward", "steps": 1}, "id": 8}\x0D\x0A
Zoom In Steps 16	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepforward", "steps": 16}, "id": 8}\x0D\x0A
Zoom Out Steps 1	{"jsonrpc": "2.0", "method": "property.set", "params": {"property": "optics.zoom.stepreverse", "steps": 1}, "id": 8}\x0D\x0A

Appendix B. Query Commands

Illumination	{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "illumination.sources.laser.power"}, "id": 2}\x0D\x0A
Input	{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "image.window.main.source"}, "id": 5}\x0D\x0A
Power	{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "system.state"}, "id": 3}\x0D\x0A
Shutter	{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "optics.shutter.target"}, "id": 4}\x0D\x0A
Test Pattern	{"jsonrpc": "2.0", "method": "property.get", "params": {"property": "image.testpattern.show"}, "id": 6}\x0D\x0A